Mathematical Theory and Methods

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Preface

THESE NOTES follow, and are meant to cement the course *Algebra*, taught at the University of Western Australia by Dr. Gordon Royle in Semester 2 of 2026. Where needed, the content of lectures is supplemented by *Linear Algebra Done Right* by Sheldon Axler.

0.1 What is Algebra?

I DO NOT YET KNOW ENOUGH TO SAY.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every *or* for all
$\exists$	There exists
$\implies$	Implies
$\in$	In *or* is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subset of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not *or* without
$\varnothing$	Empty set
$\sim$	Relates to
$\mapsto$	Maps to
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\lvert n \rvert$	The absolute value of n
$\square$	It has been proven

Part I

Linear Algebra

Chapter 1

Systems of Linear Equations

1.1 Solving Systems of Linear Equations

1.1.1 Augmented Matrices

1.1.2 Elementary Row Operations

1.1.3 Row-Echelon Form

1.2 Gauss-Jordan Elimination

1.3 Reasoning about Systems of Linear Equations

Chapter 2

Vector Spaces and Subspaces

2.1 $\mathbb{R}^n$

2.2 Vector Subspaces

2.2.1 Subspace Proofs

2.3 Span

2.4 Linear Independence

2.4.1 The Linear Independence Test

2.5 Bases, Dimension, & Coordinates

Chapter 3

Matrices and Determinants